GGG–D–691b
September 27, 1962

SUPERSEDING
Int. Fed. Spec. GGG-D-00691a (Navy-Ships)
August 11, 1959 and
Fed. Spec. GGG-D-691
September 21, 1954

FEDERAL SPECIFICATION

DRILL, MASONRY, HAND AND HOLDER, MASONRY DRILL, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers star, single-bit, twist flute, and three-flute masonry drills for manual use in drilling holes in brick, tile, concrete, stone, and other types of masonry and drill holders used with twist flute masonry drills.

1.1.1 *Federal specification coverage.* This specification does not include all types, classes, and styles of drills and drill holders covered by the title of this specification but it includes only those most commonly used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Drills and drill holders shall be of the following types, classes, and styles as specified (see 6.1):

Type I.—Star (four-flute) drill.
Type II.—Single-bit drill.
Type III.—Twist-flute drill (drill point).
  Shank style A.
  Shank style B.
  Shank style C.
Type IV.—Drill holder.
  Class 1.—Plain grip.
    Socket style A.
    Socket style B.
    Socket style C.
  Class 2.—Rubber grip.
    Socket style A.
    Socket style B.
    Socket style C.
Type V—Three-flute drill.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are

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available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of drills or drill holders which are otherwise in accordance with the requirements of this specification.

3.2 **Material.** Except for type IV, all drills shall be forged from a high quality, wear- and shock-resistant steel conforming to the chemical composition of not less than steel SAE 9255 or AISI 9255, or of a steel with equivalent or better characteristics. The drills shall be properly hardened and shall be tempered so that they will satisfactorily withstand the tests specified in 4.3.3.1 and 4.3.3.2.

3.3 **Identification marking.** Each drill shall be stamped or otherwise permanently marked with its nominal size and the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. Marking shall be in accordance with MIL-STD-130, as applicable.

3.4 Type I star (four flute) drills.

3.4.1 *Design.* The cutting end shall have four cutting edges and shall have an equal number of flutes. The cutter face shall be slightly pointed and edges shall be sharp. The striking end shall be chamfered. The flutes, lands, and cutting edges shall be concentric with the shank. The body shall be either octagonal or hexagonal in cross section.

3.4.2 *Hardness.* The cutting end shall be hardened and tempered to a Rockwell C hardness of not less than 52 nor more than 60. The striking end of drills shall have a Rockwell C hardness of not less than 32 nor more than 45 (see 4.3.1).

3.4.3 *Finish.* Star drills shall have a lacquered, enameled, or oil-treated natural finish at the option of the contractor.

3.4.4 Type I star drills shall conform to table I and shall be similar to figure 1 .

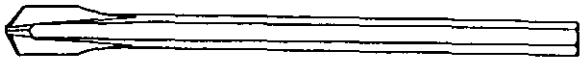


FIGURE 1.—Type I star (four fluted) drills.

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TABLE I.—*Type I star (four flute) drills*

Nominal size	Diameter cutting end		Stock size (min.)	Length fluted section (min.)	Length overall (approx.)
	Min.	Max.			
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
1/4	0.250	0.281	3/16	1-1/4	8
1/4	.250	.281	3/16	1-1/4	12
5/16	.312	.343	1/4	1-5/16	8
5/16	.312	.343	1/4	1-5/16	12
3/8	.375	.410	5/16	1-3/8	8
3/8	.375	.410	5/16	1-3/8	12
3/8	.375	.410	5/16	1-3/8	24
7/16	.437	.468	5/16	1-5/8	8
7/16	.437	.468	5/16	1-5/8	12
7/16	.437	.468	5/16	1-5/8	18
1/2	.500	.531	3/8	1-3/4	8
1/2	.500	.531	3/8	1-3/4	12
1/2	.500	.531	3/8	1-3/4	18
9/16	.562	.608	7/16	1-7/8	12
9/16	.562	.608	7/16	1-7/8	18
5/8	.625	.680	7/16	2	8
5/8	.625	.680	7/16	2	12
5/8	.625	.680	7/16	2	18
5/8	.625	.680	7/16	2	24
11/16	.687	.718	1/2	2-1/8	12
11/16	.687	.718	1/2	2-1/8	18
3/4	.750	.795	1/2	2-1/4	8
3/4	.750	.795	1/2	2-1/4	12
3/4	.750	.795	1/2	2-1/4	18
3/4	.750	.795	1/2	2-1/4	24
7/8	.875	.921	5/8	2-1/2	12
7/8	.875	.921	5/8	2-1/2	18
7/8	.875	.921	5/8	2-1/2	24
1	1.000	1.046	11/16	2-1/2	12
1	1.000	1.046	11/16	2-1/2	18
1	1.000	1.046	11/16	2-1/2	24
1	1.000	1.046	11/16	2-1/2	30
1- 1/8	1.125	1.171	3/4	2-3/4	12
1- 1/8	1.125	1.171	3/4	2-3/4	18
1- 1/4	1.250	1.296	13/16	3-1/8	12
1- 1/4	1.250	1.296	13/16	3-1/8	18
1- 1/4	1.250	1.296	13/16	3-1/8	24
1- 1/4	1.250	1.296	13/16	3-1/8	30
1- 1/4	1.210	1.296	13/16	3-1/8	36
1- 3/8	1.375	1.437	7/8	3-3/8	12
1- 3/8	1.375	1.437	7/8	3-3/8	18
1- 1/2	1.500	1.562	1	3-1/2	12
1- 1/2	1.500	1.562	1	3-1/2	18
1- 1/2	1.500	1.562	1	3-1/2	24
1- 1/2	1.500	1.562	1	3-1/2	30
1- 1/2	1.500	1.562	1	3-1/2	36
1- 3/4	1.750	1.812	1- 1/8	3-5/8	18
1- 3/4	1.750	1.812	1- 1/8	3-5/8	24
1- 3/4	1.750	1.812	1- 1/8	3-5/8	30
1- 3/4	1.750	1.812	1- 1/8	3-5/8	36
2	2.000	2.062	1- 1/8	3-3/4	18
2	2.000	2.062	—	3-3/4	24

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3.5 Type II single-bit drill.

3.5.1 *Design.* Drills shall be upset on the cutting end and forged into a double-beveled, single-cutting bit. The cutting face shall be slightly pointed and edges shall be sharp. The striking end shall be chamfered. The body shall be either octagonal or hexagonal in cross section.

3.5.2 *Hardness.* The cutting end shall be hardened and tempered to a Rockwell C hardness of not less than 52 nor more than 60. The striking end shall have a Rockwell C hardness of not less than 32 nor more than 45 (see 4.3.1).

3.5.3 *Finish.* Single-bit drills shall have a lacquered, enameled, or oil-treated natural finish at the option of the contractor.

3.5.4 Type II single-bit drills shall conform to table II and shall be similar to figure 2.



FIGURE 2.—Type II single-bit drills.

TABLE II.—*Type II single-bit drills*

Nominal size	Width cutting end		Stock size (min.)	Length overall (approx.)
	Min.	Max.		
Inches	*Inches*	*Inches*	*Inch*	*Inches*
7/16	7/16	15/32	5/16	18
1/2	1/2	17/32	3/8	12
1/2	1/2	17/32	3/8	18
5/8	5/8	21/32	7/16	12
5/8	5/8	21/32	7/16	18
3/4	3/4	25/32	1/2	12
3/4	3/4	25/32	1/2	18
3/4	3/4	25/32	1/2	30
3/4	3/4	25/32	1/2	48
3/4	3/4	25/32	1/2	72
7/8	7/8	29/32	5/8	12
7/8	7/8	29/32	5/8	18
1	1	1- 1/32	3/4	12
1	1	1- 1/32	3/4	18
1	1	1- 1/32	3/4	30
1	1	1- 1/32	3/4	48
1-1/8	1-1/8	1- 5/32	3/4	12
1-1/8	1-1/8	1- 5/32	3/4	18
1-1/8	1-1/8	1- 5/32	3/4	72
1-1/4	1-1/4	1- 9/32	7/8	12
1-1/4	1-1/4	1- 9/32	7/8	18
1-1/4	1-1/4	1- 9/32	7/8	24
1-1/4	1-1/4	1- 9/32	7/8	30
1-1/4	1-1/4	1- 9/32	7/8	48
1-3/8	1-3/8	1-13/32	7/8	18
1-1/2	1-1/2	1-17/32	1	18
1-1/2	1-1/2	1-17/32	1	72
1-3/4	1-3/4	1-25/32	1	18
1-3/4	1-3/4	1-25/32	1	22
1-3/4	1-3/4	1-25/32	1	30
1-3/4	1-3/4	1-25/32	1	48
1-3/4	1-3/4	1-25/32	1	72
2	2	2- 1/32	1	18

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3.6 Type III. twist-flute drills (drill point) (with tapered shank). The twist-flute drills shall be provided with a tapered shank as specified hereinafter for each shank size. The drills shall be provided with either two or three right-hand twist flutes.

3.6.1 *Material.* Drills shall be machined or forged from solid bars.

3.6.2 *Cutting edge.* The cutting edge of the drill shall be suitably ground, ready for use and be capable of being reground when necessary.

3.6.3 *Hardness.* The cutting edge and the full length of the flutes of the drills shall be hardened and tempered to show not less than 55 or more than 64 on a Rockwell C scale (see 4.3.1).

3.6.4 *Finish.* Drills shall be free from burrs, roughness, and excessive oxide scale and shall have an oil-treated natural finish unless otherwise specified (see 6.1). The shank shall be smoothly finished and free from pits, nodules, or other surface blemishes. The small end of the shank shall be either chamfered or rounded.

3.6.4 1 *Shank style A.* The drill shall be provided with a shank conforming to figure 3 and be suitable for use with type IV, classes 1 and 2, socket style A drill holder. The drill shall conform to table III.

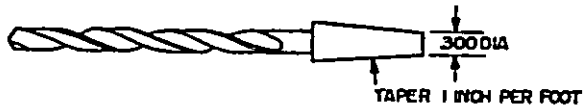


FIGURE 3.—Type III, shank style A, twist-flute drill (drill point).

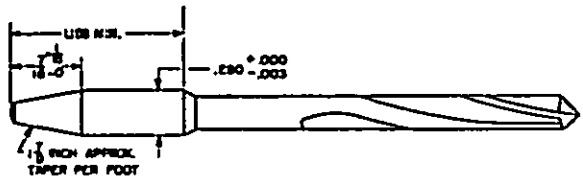


FIGURE 4.—Type III, shank style B, twist-flute drill (drill point)

3.6.4 2 *Shank style B.* The drill shall be provided with a shank conforming to figure 4 and shall be suitable for use with type IV, classes, 1 and 2, socket style B drill holders. The small end of the shank shall be either spherical or have the corners radiused. The drill shall conform to table IV.

TABLE III.—*Type III, shank style A, twist-flute drill (drill point)*

Drill size			Overall length +1/2, —1/8 inch	Drilling depth (minimum)
Nominal diameter	Minimum diameter	Maximum diameter		
Inch	*Inch*	*Inch*	*Inches*	*Inches*
11/64	0.174	0.181	2-5/8	1-1/4
3/16	.198	.213	2-5/8	1-5/16
7/32	.220	.257	3-1/16	1-5/8
1/4	.265	.282	3-1/4	1-3/4
9/32	.286	.302	3-5/16	1-15/16
5/16	.312	.334	4-1/4	2-3/8
11/32	.342	.355	4-5/16	2-3/8
3/8	.371	.410	5	3
7/16	.450	.472	5-1/8	3-3/16
1/2	.527	.549	5-9/16	3-3/16
9/16	.585	.604	5-9/16	3-3/16
5/8	.655	.665	6	3-3/16
3/4	.780	.790	6	3-3/16

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TABLE IV.—*Type III, shank style B, twisted-flute (drill point)*

For fibre plug size	Drill point dimensions			
	Drill size		Length of flutes ±1/8 inch	Overall length ±1/8 inch
	Min.	Max.		
Number	*Inch*	*Inch*	*Inches*	*Inches*
6	0.145	0.152	1	2-25/32
8	.175	.183	1-5/8	3-11/32
10	.198	.213	1-5/8	3-11/32
12	.247	.257	2	3-27/32
14	.272	.282	2-3/16	3-27/32
16	.325	.335	2-5/8	4- 3/8
20	.393	.403	2-5/8	4- 5/8

3.6.4 3 *Shank style C.* The drill shall be provided with a shank conforming to figure 5 and be suitable for use with type IV, classes 1 and 2, socket style C drill holders. The small end of the shank shall be either spherical or have the corners radiused. The drill shall conform to table V.

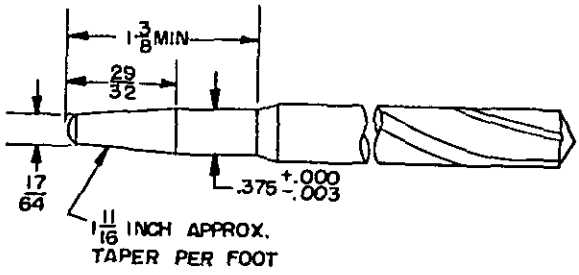


FIGURE 5.—Type III, shank style C, twist-flute drill (drill point).

3.7 Type IV, drill holder. Type IV drill holder shall be provided with a tapered socket to accommodate the shank of type III drill. A slot shall be provided in the socket end of the holder to facilitate the removal of the drill from the holder by an ejector. The striking end of the holder shall be chamfered to eliminate sharp corners and, when tested as specified in 4.3.1, shall have a hardness of not less than 32 nor more than 48 on a Rockwell C scale.

3.7.1 *Ejector.* The drill ejector shall be of steel, shaped in a manner to engage the holder and eject the drill. The ejector shall be properly heat-treated to preclude the possibility of spalling, burring or deforming under normal use.

3.7.2 *Class 1, plain grip.* Class 1 drill holder shall be of steel, hexagonal or octagonal in cross section. Class 1 drill holder shall be similar to figure 6.

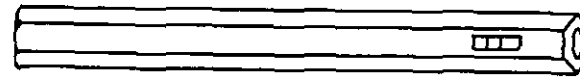


FIGURE 6.—Type IV, class 1 drill holder.

TABLE V.—*Type III, shank style C, twist-flu te drill (drill point)*

For fibre plug size	Drill point dimensions			
	Drill size		Length of flute ±1/16 inch	Overall length ±1/8 inch
	Min.	Max.		
Number	*Inch*	*Inch*	*Inches*	*Inches*
16	0.312	0.321	2- 7/16	4-3/8
20	.371	.375	3- 1/8	5-1/8
7/16	.468	.472	3- 1/8	5-1/4
1/2	.545	.549	3- 3/8	5-7/16
5/8	.656	.660	3- 3/4	6
3/4	.781	.785	3-23/32	6

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3.7.2.1 *Socket style A*. Socket style A drill holder shall be suitable for use with the shank style A drill (type III) and shall have a socket tapering one inch per foot to suitably engage all sizes specified in table III.

3.7.2 2 *Socket style B*. Socket style B drill holder shall be suitable for use with the shank style B drill (type III) and shall have a socket tapering approximately 1-7/8 inch per foot to suitably engage all sizes specified in table IV.

3.7.2 3 *Socket style C*. Socket style C drill holders shall be suitable for use with the shank style C drill and shall have a socket tapering approximately 1-11/16 inch per foot to suitably engage all sizes specified in table V.

3.7.3 *Class 2 rubber grip*. Class 2 drill holder shall be of steel and shall be provided with a molded or otherwise firmly attached rubber handgrip of good quality hard rubber. The rubber grip shall be flanged on the end nearest the striking end of the holder to prevent injury to the user's hand. The rubber grip shall not loosen under service conditions. Class 2 holder shall be similar to figure 7.

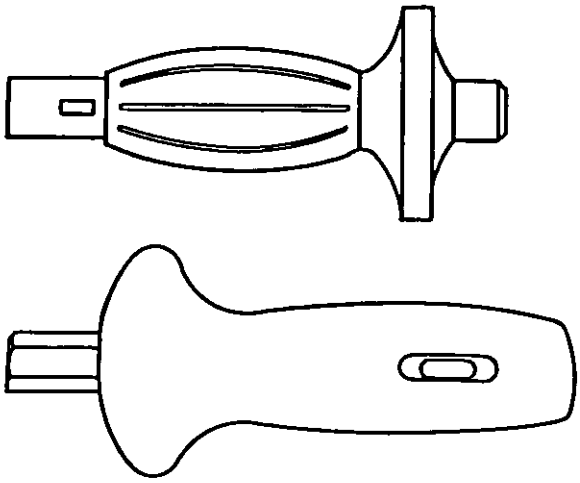


FIGURE 7.—Type IV, class 2 drill holder.

3.7.3.1 *Socket style A*. Socket style A shall be as specified in 3.7.2.1.

3.7.3 2 *Socket style B*. Socket style B shall be as specified in 3.7.2.2.

3.7.3 3 *Socket style C*. Socket style C shall be as specified in 3.7.2.3.

3.8 Type V three-flute drills.

3.8.1 *Design*. The cutting end shall have three cutting edges and an equal number of flutes. The cutting face shall be slightly pointed and edges shall be sharp. The striking end shall be chamfered. The flutes, lands, and cutting edges shall be concentric with the shank. The body shall be either octagonal or hexagonal in cross section.

3.8.2 *Hardness*. The cutting end shall be hardened and tempered to a Rockwell C hardness of not less than 52 nor more than 60. The striking end of drill shall have a Rockwell C hardness of not less than 32 nor more than 45 (see 4.3.1).

3.8.3 *Finish*. Drills shall have a lacquered, enameled, or oil-treated natural finish at the option of the contractor.

3.8.4 Type V drills shall conform to table VI and be similar to figure 8.

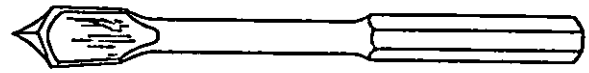


FIGURE 8.—Type V three-flute drills.

3.9 Workmanship. Workmanship shall be first class in every respect. Drills and drill holders shall be free from rust, laps, cracks, and other defects which may impair serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services ac-

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ceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Acceptance inspection.**

4.2.1 *Lot.* All drills and drill holders of the same class, type, style, and size shall be considered a lot for purposes of acceptance inspection.

4.2.2 *Examination.* A sample of drills and holders shall be selected at random as specified in table VII from each inspection lot of material for examination, with lot acceptance based on the sampling inspection requirements of table VII in accordance with the procedures of MIL-STD-105. Each of the sample drills and drill holders shall be examined to verify compliance with this specification. Defects shall be classified as specified in table VIII. Any drill or drill holder in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE *VI.—Type V three-flute drills*

Nominal size	Overall handle length (min.)	Diameter of cutting end		Stock size (min.)	Length of flute	Length overall
		Nominal (min.)	Max.			
Inches	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inches*
3/8	5	3/8	27/64	9/16	1-3/4	9
7/16	5	7/16	15/32	9/16	2	10
7/16	5	7/16	15/32	9/16	2	12
1/2	5	1/2	35/64	9/16	2	10
1/2	5	1/2	35/64	9/16	2	12
9/16	5-1/8	9/16	39/64	5/8	2	12
9/16	5-1/8	9/16	39/64	5/8	2	18
5/8	5-1/8	5/8	43/64	5/8	2	12
5/8	5-1/8	5/8	43/64	5/8	2	18
11/16	5-1/8	11/16	47/64	5/8	2-1/8	12
11/16	5-1/8	11/16	47/64	5/8	2-1/8	18
3/4	5-1/8	3/4	51/64	11/16	2-1/4	12
3/4	5-1/8	3/4	51/64	11/16	2-1/4	18
13/16	5-1/8	13/16	55/64	11/16	2-1/2	12
13/16	5-1/8	13/16	55/64	11/16	2-1/2	18
7/8	5-1/8	7/8	59/64	11/16	2-1/2	12
7/8	5-1/8	7/8	59/64	11/16	2-1/2	18
1	5-1/8	1	1- 3/64	3/4	2-1/2	12
1	5-1/8	1	1- 3/64	3/4	2-1/2	18
1- 1/8	5-1/8	1- 1/8	1-11/64	3/4	3	12
1- 1/8	5-1/8	1- 1/8	1-11/64	3/4	3	18
1- 1/4	9	1- 1/4	1- 5/16	3/4	3	12
1- 1/4	9	1- 1/4	1- 5/16	3/4	3	18
1- 3/8	9	1- 3/8	1- 7/16	3/4	3	12
1- 3/8	9	1- 3/8	1- 7/16	3/4	3	18
1- 1/2	8-1/2	1- 1/2	1- 9/16	7/8	3-1/2	12
1- 1/2	8-1/2	1- 1/2	1- 9/16	7/8	3-1/2	18
1- 5/8	8-1/2	1- 5/8	1- 3/4	7/8	3-1/2	12
1- 5/8	8-1/2	1- 5/8	1- 3/4	7/8	3-1/2	18

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TABLE VII.—*Sampling for visual and dimensional examination AQL (approx.)=1.5 percent defective*

Number of drills or drill holders in inspection lot	Sample size, number to be examined (reference Standard MIL-STD-105)
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

TABLE VIII.—*Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, style, and size (as applicable) not as specified.
102	Material defective; evidence of deep pits, cracks, rust, laps, or foreign inclusions.
103	Axis of drill not straight; axis or surface of holder not straight.
104	Flutes of drills not clean and well defined.
105	Cutting face of drills not pointed or cutting edges not sharp; evidence of nicks or burrs.
106	Shriking end not chamfered (except type III).
107	Flutes, lands, and cutting edges not concentric with the shank (type I and V drills only).
108	Diameter of shank at small end not as required; taper of shank not as specified (type III only).
109	Cutting end of drill not double-beveled or not upset (type II only).
110	Width (or diameter) of cutting end not within the specified tolerances (types I, II, and V only).
111	Stock size less than the specified minimum (types I, II, and V only).
112	Rubber handgrip defective, not secure, or not flanged (type IV only).
113	Taper socket not as required (to accept the drill) (type IV only).
Minor:	
201	Length overall not as required.
202	Length of fluted section less than specified (type I only).
203	Drilling depth less than specified (type III only).
204	Finish defective or incomplete; evidence of flaking, blisters, or uncoated areas.
205	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.2.3 *Acceptance tests.* A sample of drills or drill holders shall be selected at random from each inspection lot in accordance with table IX and shall be subjected to the hardness, drilling, and marking tests specified in 4.3.1 and 4.3.3 to verify compliance with this specification. Any sample item which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE IX.—*Sampling for lot acceptance tests*

Number of drills or drill holders in inspection lot	Number of drills or drill holders in sample	Number of items nonconforming in all tests	
		Acceptance number	Rejection number
40 or under .	5	0	1
41 to 110 ...	7	0	1
111 to 500 ...	10	0	1
501 to 1300 ..	15	1	2
1301 and over .	25	1	2

4.3 Test procedures.

4.3.1 *Hardness test.* Hardness tests shall be conducted to determine conformance with the applicable requirements specified in 3.4.2, 3.5.2, 3.6.3, 3.7 and 3.8.2. Impressions shall be taken within a distance of 1/4 to 1/2 inch back of the cutting edge and within 1/2 inch of the striking end.

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4.3.2 *Chemical analysis tests.* Chemical analysis tests shall be conducted in accordance with Fed. Test Method Std. No. 151, or a certificate of analysis may be accepted if satisfactory to the procuring activity (see 3.2).

4.3.3 *Drilling and marking tests.*

4.3.3.1 *Drilling test.* The drill shall be supported manually or by other suitable means, at the discretion of the inspector. With the cutting end against a solid concrete block the striking end of the drill or drill holder, as applicable, shall be struck with a hammer of suitable weight as specified below until a hole of not less than 1/2 inch in depth has been attained. A minimum of three holes shall be drilled.

For drill sizes 3/16 through 7/16 inclusive, use 1-pound hammer.

For drill sizes 1/2 through 5/8 inclusive, use 1-1/2-pound hammer.

For drill sizes 11/16 through 1 inclusive, use 2-pound hammer.

For drill sizes 1-1/8 through 2 inclusive, use 3-pound hammer.

4.3.3.2 *Marking test.* Following the drilling test, drills and drill holders shall be subjected to the marking test Using a hammer of the proper weight as specified in 4.3.3.1, three impressions shall be made with the cutting end of the drill on a steel plate having a Rockwell C hardness of not less than 37 nor more than 40.

4.3.3.3 When tested as specified in 4.3.3.1 and 4.3.3.2, the striking end of the drill or drill holder shall not mushroom to the extent that cracks develop on the edge. There shall be no indication of bending, deforming, or fracture in the body of the drill, nor any signs of spalling, chipping, or cracking of the cutting end. Failure of drill to perform satisfactorily or failure otherwise to pass the above specified tests shall be cause for rejection.

4.4 Inspection of preparation for delivery. Sample articles, packages, and packs shall be selected and inspected to verify conformance to section 5 herein.

5. PREPARATION FOR DELIVERY

Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Drills and drill holders shall be preserved and packaged in such a manner that will afford adequate protection against corrosion, deterioriation, and physical damage during shipment from supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with MIL-H-15424.

5.2.3 *Level C.* Drills and drill holders, preserved and packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 *Military agencies.* In addition to any special marking specified in the contract or order, or herein, interior packages and shipping containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and shank or socket style, if applicable (see 1.2, 3.7.2.1, 3.7.2.2 and 3.7.2.3).
(c) Size (nominal) and length (see applicable table).
(d) If other than finish specified for type III is required (see 3.6.4).
(e) Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
(f) Marking, if different (see 5.3).

6.2 Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—O
Navy—Sh
Air Force—MOA

Copies of this specification may be purchased for 10 cents each.

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